Answer Key - Physics - Undergraduate Level Assessment 20 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** D
   Options: A) Diode laser; B) Dye laser; C) Free-electron laser; D) Gas laser
   Note: Correct option: D - Gas laser

2. **Answer:** A
   Options: A) 4; B) 5; C) 6; D) 20
   Note: Correct option: A - 4

3. **Answer:** D
   Options: A) 0.25 W; B) 0.5 W; C) 1 W; D) 4 W
   Note: Correct option: D - 4 W

4. **Answer:** D
   Options: A) 1; B) 2; C) 3; D) 5
   Note: Correct option: D - 5

5. **Answer:** B
   Options: A) Constant temperature; B) Constant volume; C) Constant pressure; D) Adiabatic
   Note: Correct option: B - Constant volume

True / False

6. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: 'interactions with the ionic lattice'.

7. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

8. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

9. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

10. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** I and II. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 0.25 mc^2. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key